

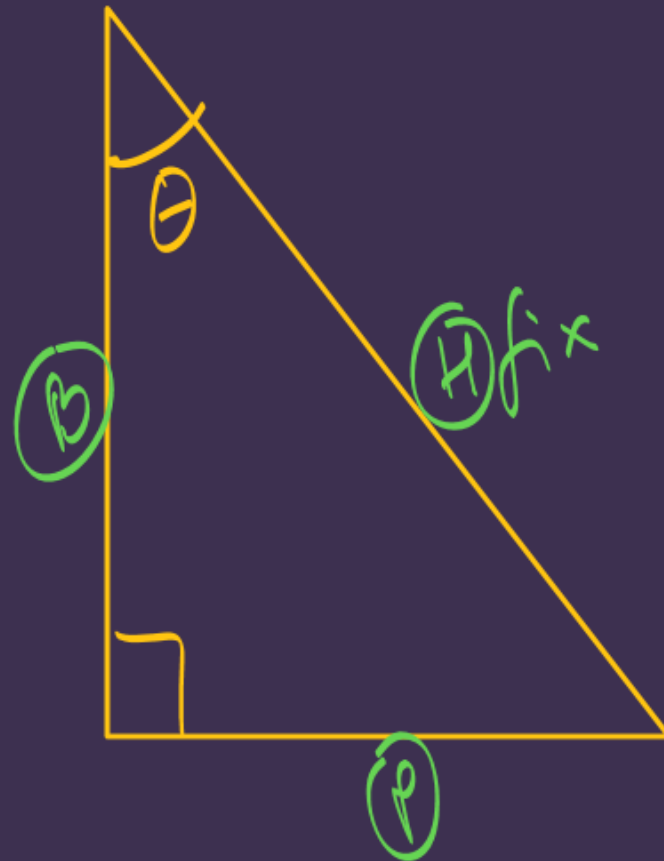
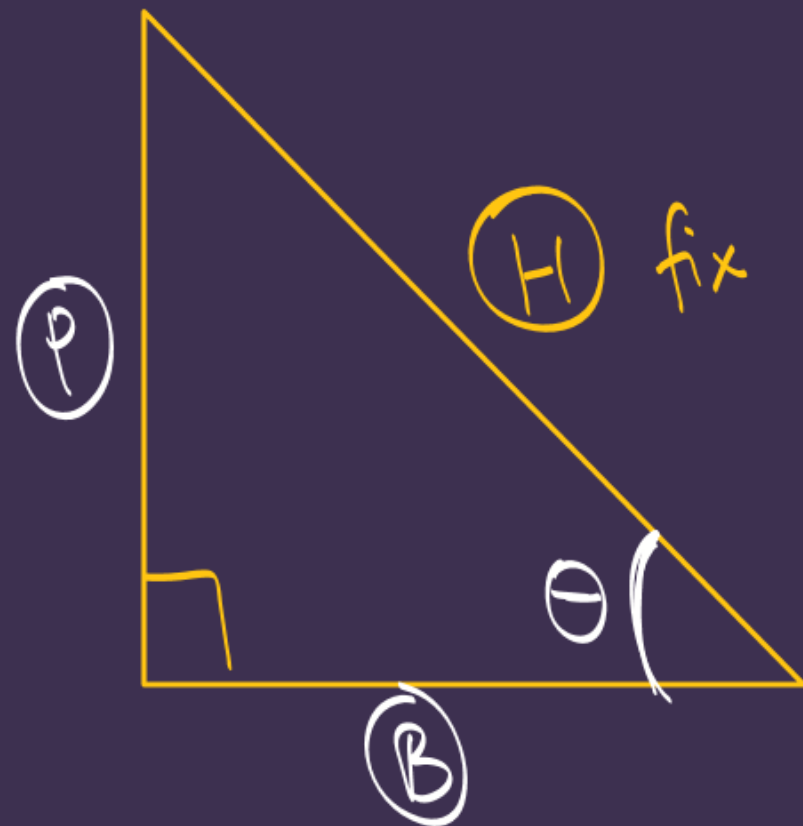
[Aim: 100/100 in Maths]



TRIGONOMETRY

Lecture #2

#ATK²:-



$\sin \theta$	$\cos \theta$	$\tan \theta$
P	B	P
H	H	B
$\csc \theta$	$\sec \theta$	$\cot \theta$

~~K³B~~

$$\sin \theta \neq \sin \times \theta$$

read as "sin of angle θ "

$$\sin(A+B) \neq \sin A + \sin B$$

$$(\cos \theta)^2 = \cos^2 \theta \checkmark$$

$$(\sec \theta)^3 = \sec^3 \theta \checkmark$$

Unknown side $\Rightarrow$ Pytha $\Rightarrow$

$$H^2 = P^2 + B^2$$

बिना एक T-ratio के बाकी और T-ratios पता चल सकते हैं!!

#LP: If $\sin A = \frac{3}{5}$, find all other trigonometric ratios.

P B P
H H B

$$\sin A = \frac{3}{5}$$

$$\frac{P}{H} = \frac{3}{5}$$

Let $\begin{matrix} P = 3k \\ H = 5k \end{matrix}$ B??

by pyth $H^2 = P^2 + B^2$

$$(5k)^2 = (3k)^2 + B^2$$

$$25k^2 = 9k^2 + B^2$$

$$25k^2 - 9k^2 = B^2$$

$$B^2 = 16k^2 \Rightarrow B = \sqrt{16k^2} \Rightarrow 4k$$

$$\begin{matrix} P = 3k \\ H = 5k \\ B = 4k \end{matrix}$$

To find

$$\cos A = \frac{B}{H} \Rightarrow \frac{4k}{5k} \Rightarrow \frac{4}{5}$$

$$\tan A = \frac{P}{B} \Rightarrow \frac{3k}{4k} \Rightarrow \frac{3}{4}$$

$$\operatorname{cosec} A = \frac{H}{P} = \frac{5k}{3k} \Rightarrow \frac{5}{3}$$

$\rightarrow \sin A \text{ का 3 मल } \Rightarrow \frac{5}{3}$

$$\sec A = \frac{H}{B} = \frac{1}{\cos A} \Rightarrow \frac{5}{4}$$

$$\cot A = \frac{B}{P} = \frac{1}{\tan A} \Rightarrow \frac{4}{3}$$

iii) find the value of the expression $\Rightarrow \frac{\tan A + \sin A}{1 - \cot A}$

(iii) Find value of $\frac{\sin A}{\cot A}$

$$\Rightarrow \frac{\left(\frac{3}{5}\right)}{\left(\frac{4}{3}\right)} \Rightarrow \frac{3}{5} \times \frac{3}{4} \Rightarrow \frac{9}{20}$$

$$\Rightarrow \frac{\left(\frac{3}{4}\right) + \left(\frac{3}{5}\right)}{1 - \left(\frac{4}{3}\right)}$$

$$\Rightarrow \frac{\frac{15 + 12}{20}}{\frac{3 - 4}{3}} \Rightarrow \frac{\frac{27}{20}}{-\frac{1}{3}}$$

$$\Rightarrow \frac{27}{20} \times -\frac{3}{1} \Rightarrow -\frac{81}{20}$$

#LP: If $\operatorname{cosec}\theta = \sqrt{10}$, find the values of all T-ratios of θ .

#LP: If $\sin A = \frac{1}{3}$ then $\cos A \cdot \operatorname{cosec} A + \tan A \cdot \sec A =$

P B P
H H B

$$\sin A = \frac{1}{3}$$

$$\frac{P}{H} = \frac{1}{3}$$

$$P = 1k, H = 3k$$

Pytho

$$H^2 = P^2 + B^2$$

$$(3k)^2 = (k)^2 + B^2$$

$$9k^2 - k^2 = B^2$$

$$B^2 = 8k^2 \rightarrow B = \sqrt{8k^2} \rightarrow 2\sqrt{2}k$$

$$H = 3k$$

$$P = k$$

$$B = 2\sqrt{2}k$$

$$\Rightarrow \left(\frac{B}{H} \right) \times \left(\frac{H}{P} \right) + \left(\frac{P}{B} \right) \times \left(\frac{H}{B} \right)$$

$$\Rightarrow \frac{2\sqrt{2}k}{k} + \left(\frac{k}{2\sqrt{2}k} \right) \times \left(\frac{3k}{2\sqrt{2}k} \right)$$

$$\Rightarrow 2\sqrt{2} + \left(\frac{1}{2\sqrt{2}} \right) \left(\frac{3}{2\sqrt{2}} \right)$$

$$\Rightarrow 2\sqrt{2} + \frac{3}{4 \times 2}$$

$$\Rightarrow 2\sqrt{2} + \frac{3}{8} \Rightarrow \frac{16\sqrt{2} + 3}{8} k$$

$$\begin{array}{r|l} 2 & 8 \\ \hline 2 & 4 \\ \hline 2 & 2 \\ \hline & 1 \end{array}$$

$$\sqrt{8} = 2\sqrt{2} \quad \text{How?}$$

$$\sqrt{8} = \sqrt{\underline{2 \times 2} \times \textcircled{2}}$$

$$\sqrt{8} = 2\sqrt{2}$$

#LP: If $5 \tan \alpha = 4$, show that $\frac{5 \sin \alpha - 3 \cos \alpha}{5 \sin \alpha + 2 \cos \alpha} = \frac{1}{6}$

$$5 \tan \alpha = 4$$

$$\tan \alpha = \frac{4}{5}$$

$$\frac{P}{B} = \frac{4}{5}$$

$$\text{let } P = 4k, B = 5k$$

by pytha

$$H^2 = P^2 + B^2$$

$$H^2 = (4k)^2 + (5k)^2$$

$$H^2 = 16k^2 + 25k^2$$

$$H^2 = 41k^2 \Rightarrow H = \sqrt{41}k$$

$$\begin{aligned} P &= 4k \\ B &= 5k \\ H &= \sqrt{41}k \end{aligned}$$

$$\text{LHS} \rightarrow \frac{5 \sin \alpha - 3 \cos \alpha}{5 \sin \alpha + 2 \cos \alpha}$$

$$\Rightarrow \frac{5 \times \left(\frac{P}{H}\right) - 3 \left(\frac{B}{H}\right)}{5 \times \left(\frac{P}{H}\right) + 2 \left(\frac{B}{H}\right)}$$

$$\rightarrow \frac{5 \times \left(\frac{4k}{\sqrt{41}k}\right) - 3 \left(\frac{5k}{\sqrt{41}k}\right)}{5 \left(\frac{4k}{\sqrt{41}k}\right) + 2 \left(\frac{5k}{\sqrt{41}k}\right)} \Rightarrow \frac{20}{\sqrt{41}} - \frac{15}{\sqrt{41}}$$

$$\frac{5}{6} = \frac{H}{H} \times \frac{\sqrt{41}}{\sqrt{41}} \Rightarrow \frac{5}{6} = \frac{H}{H}$$

$$\frac{5}{6} = \frac{H}{H}$$

$$\frac{20-15}{\sqrt{41}} = \frac{5}{\sqrt{41}}$$

$$\begin{matrix} P & B & P \\ H & H & B \end{matrix}$$

#LP: In a right-angled triangle ABC, right-angled at B, if AB = 12 and BC = 5, find:

(i) sin A and tan A

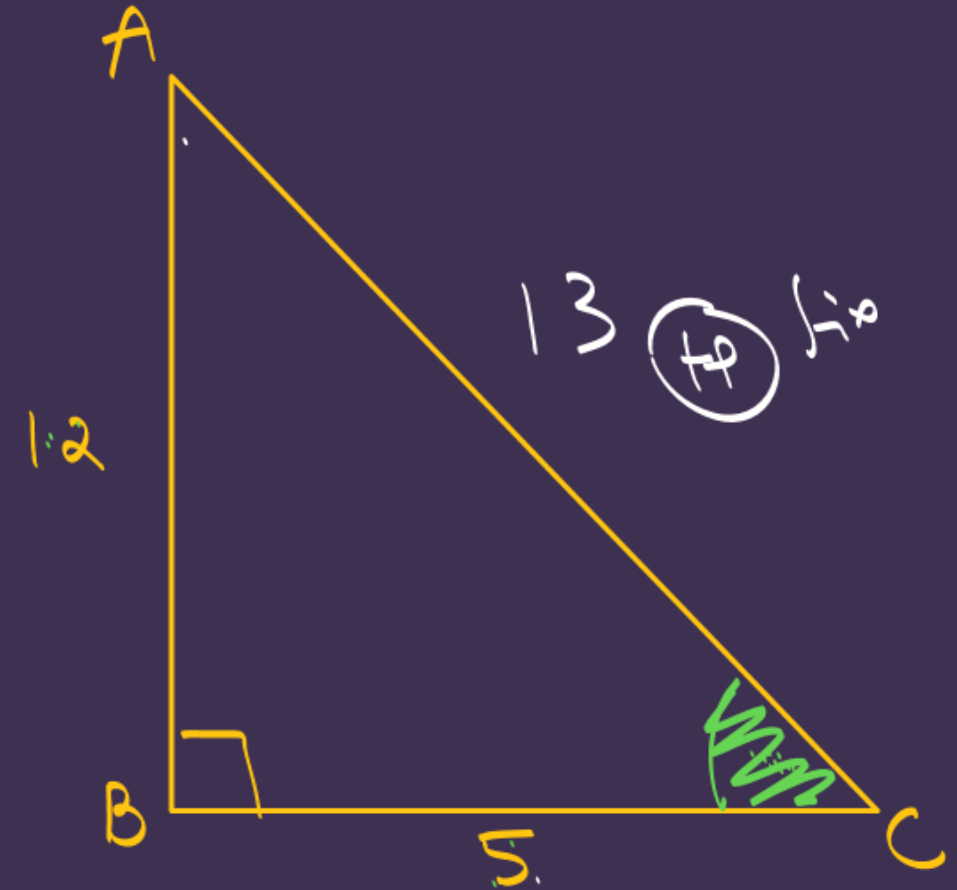
(ii) cos C and cot C

$$\textcircled{i} \sin A = \frac{P}{H} = \frac{5}{13}$$

$$\tan A = \frac{P}{B} = \frac{5}{12}$$

$$\textcircled{ii} \cos C = \frac{B}{H} = \frac{5}{13} \checkmark$$

$$\cot C = \frac{B}{P} \Rightarrow \frac{5}{12} \checkmark$$



Pytho

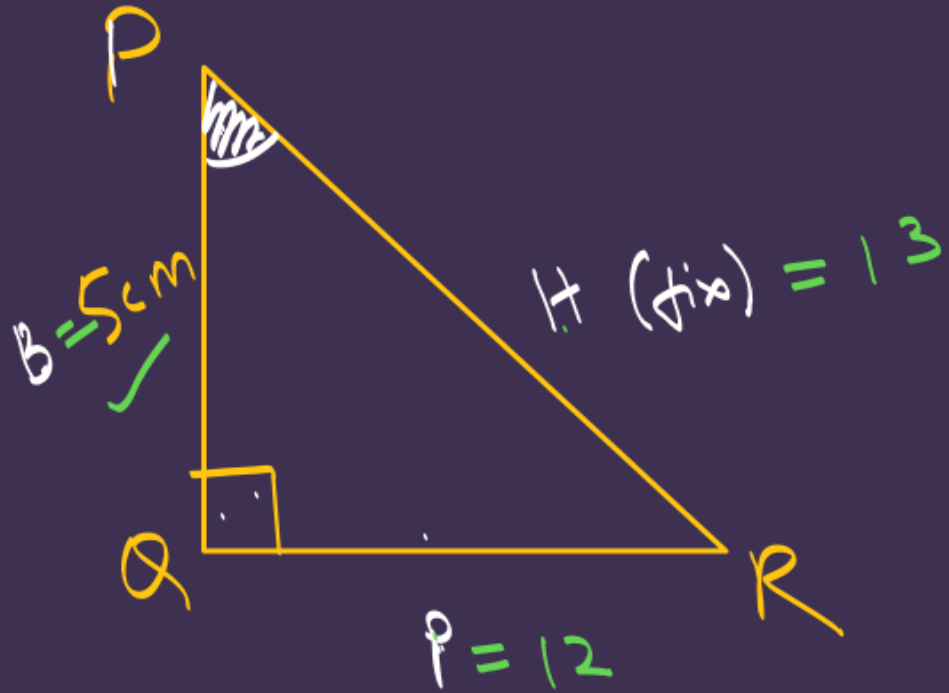
$$H^2 = P^2 + B^2$$

$$H^2 = (12)^2 + (5)^2$$

$$H^2 = 144 + 25$$

$$H = \sqrt{169} = 13 \checkmark$$

#LP: In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Determine the values of $\sin P$, $\cos P$ and $\tan P$.



Unknown sides

Pytha

given $\Rightarrow PR + QR = 25$ — (I)

$$H^2 = p^2 + b^2$$

$$(PR)^2 = QR^2 + (5)^2$$

$$PR^2 - QR^2 = 25$$

$$(PR + QR)(PR - QR) = 25$$

$$(25)(PR - QR) = 25$$

$$PR - QR = 1$$
 — (II)

$$PR = 13$$

$$QR = 12$$

To find

$$\sin P = \frac{p}{H} = \frac{12}{13}$$

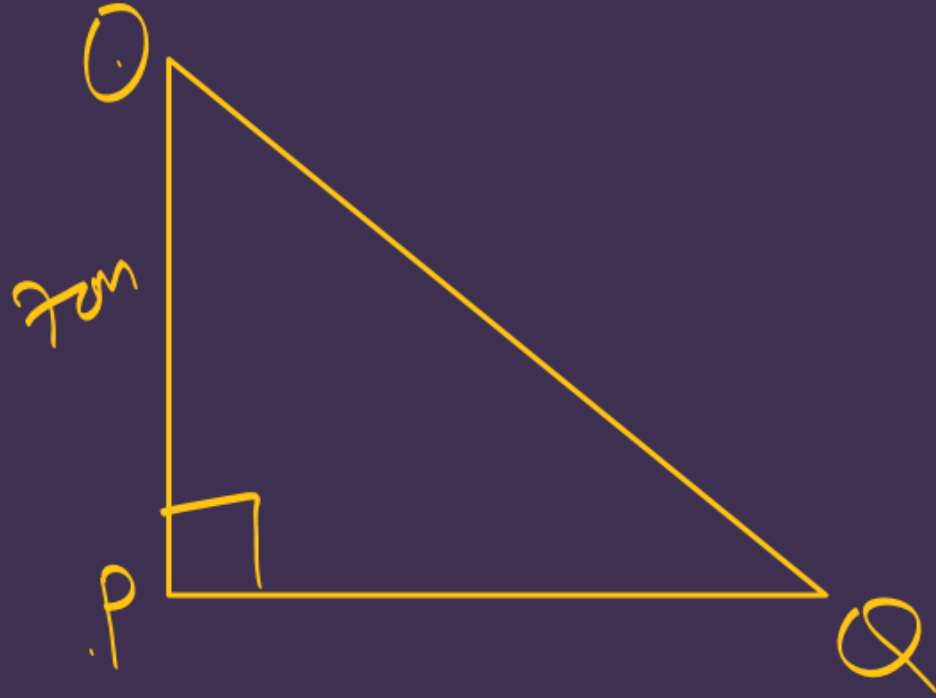
$$\cos P = \frac{b}{H} = \frac{5}{13}$$

$$\tan P = \frac{p}{b} = \frac{12}{5}$$

Ans

#LP: In $\triangle OPQ$ right-angled at P, $OP = 7$ cm, $OQ - PQ = 1$ cm. Find $\sin Q$ and $\cos Q$

$$OQ - PQ = 1$$



$$\sin Q =$$

$$\cos Q =$$

#LP: Express the ratios $\cos A$, $\tan A$ and $\sec A$ in terms of $\sin A$.

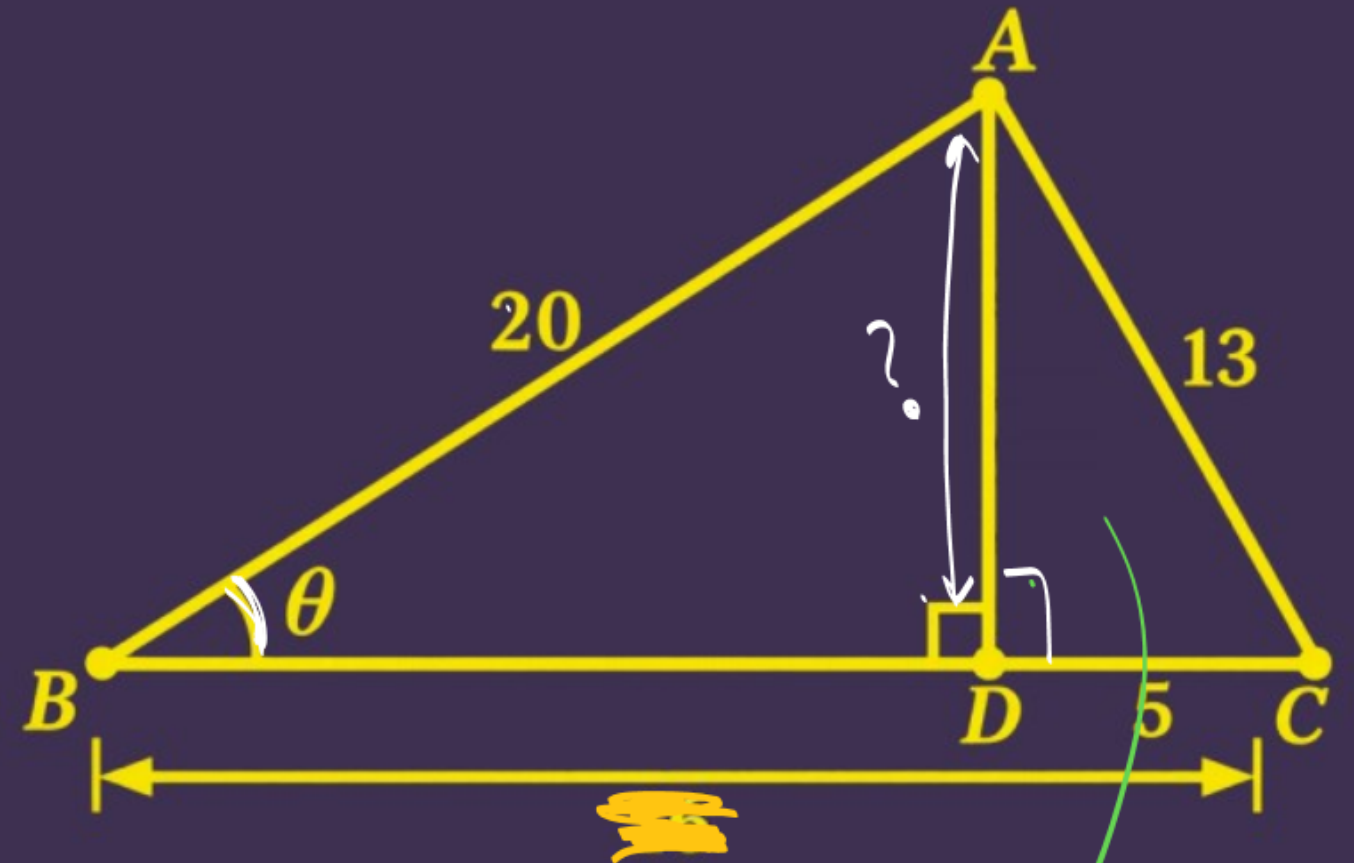
#LP: Express the trigonometry ratios $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.

#LP: Find the value of $\sin\theta$

$$\sin\theta = \frac{P}{H} = \frac{AD}{20}$$

$$= \frac{12}{20}$$

$$\sin\theta = \frac{3}{5}$$



Pytha

$$(13)^2 = AD^2 + (5)^2$$

$$169 - 25 = AD^2$$

$$AD^2 = 144$$

$$AD = \sqrt{144} = 12 \text{ cm}$$

THANK YOU COODIESS !!

